

Quantitative Aptitude

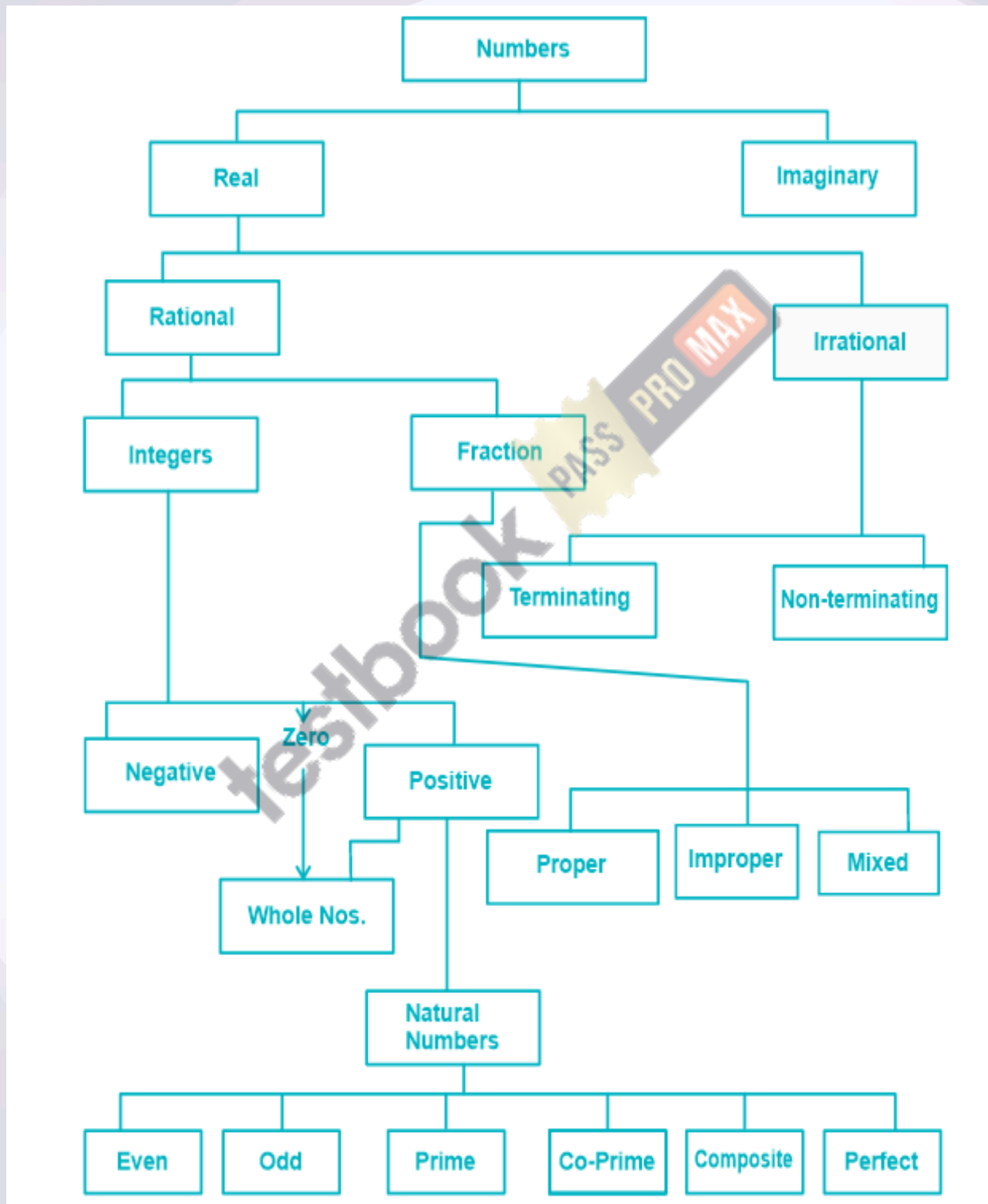
Number System

STUDY NOTES

Classification of Number System



Number System



Real Number:

The set of all rational numbers and irrational numbers are called Real Numbers. Ex; 1, $\frac{2}{3}$, 0.6, $\sqrt{3}$, π , e etc.

- The sum or difference between a rational and an irrational number is irrational.
- Ex:- $(5 + \sqrt{2})$, $(\frac{5}{2} - \sqrt{3})$ etc
- The Product of a rational and an irrational number is irrational.
- Ex:- $4\sqrt{2}$, $-5\sqrt{2}$ etc

Example 1: Which of the following is an example of a real number?

1. $\sqrt{4}$
2. $\sqrt{-9}$
3. $\frac{22}{7}$
4. Both 1 and 3.

Answer: 4 (Both $\sqrt{4}$ and $\frac{22}{7}$ are real numbers, $\sqrt{-9}$ is imaginary number)

Example 2: Which of the following statements is true regarding real numbers?

1. All integers are real numbers.
2. All fractions are irrational numbers.
3. π is a rational number.
4. -2 is not a real number.

Answer: 1 (All integers are real numbers.)

Example 3: The product of any two real numbers is:

1. Always a real number
2. Always a rational number

3. Always an irrational number
4. Sometimes an irrational number

Answer: 1 (The product of any two real numbers is always a real number.)

Imaginary Number:

A number of the form $(a + ib)$ is called a complex number where a and b are any real number and i is called iota ($\sqrt{-1}$). Ex; $\sqrt{-2} = 2i$, $2 + 3\sqrt{-5} = 2 + i5$, etc.

Example 4: Which of the following numbers is an irrational number?

1. 2
2. $7i$
3. $\sqrt{3} - 3$
4. $2 / 3$

Solution:

Concept Used:

1. Real numbers-

- Real numbers can be divided into rational numbers and irrational numbers.

2. Rational numbers-

- A rational number is a type of real number that is in the form of p/q where q is not equal to zero.

3. Irrational numbers-

- Irrational numbers are real numbers that cannot be represented as simple fractions.
- An irrational number cannot be expressed as a ratio, such as p/q , where p and q are integers, $q \neq 0$.

Explanation:

Option 02-

- $7i$ - This is not a real number, as i ($i = \sqrt{-1}$) is an imaginary number.
- Thus, we can't place it under the rational or irrational category.

Option 03-

- $(\sqrt{3} - 3)$ - Here $\sqrt{3}$ is a real number, and it can't be expressed in the form of p/q .
- Thus, $(\sqrt{3} - 3)$ is an irrational number.

∴ The correct answer is Option 3.

Rational Number:

A number of the form p/q where p and q are whole numbers and q does not equal to zero and it is denoted by Q .

Ex; $1/2$, $2/3$, $-5/2$, 0 , 2 , 5

- It can be negative, positive and also zero because we can write 0 as $0/1$.

Example 5: What is the sum of a rational number and its additive inverse?

Solution:

Calculation:

Let's assume that rational number = a

Additive inverse = $-a$

$$\Rightarrow \text{Sum} = a + (-a) = 0$$

∴ The sum of a rational number and its additive inverse is '0'.

∴ The correct answer is 0.



The sum of a rational number and its additive inverse is always '0'.

Irrational Number:

Numbers which can neither be written as a recurring decimal nor in repeating decimal or (A number that can't be written in the form of p/q). Ex; $\sqrt{2}$, $\sqrt{3}$, π , e etc.

Characteristics of Irrational Numbers:

- Irrational numbers consist of endless, non-repeating decimal expansions.
- They can be depicted on a number line.
- The sum of an irrational and a rational number is irrational.
- Operations such as addition, subtraction, multiplication, and division on irrational numbers may yield rational or irrational results.
- Multiplying an irrational number by any non-zero rational number results in an irrational number.
- Often referred to as surds, these include expressions with roots such as square roots or cube roots. While all surds are irrational, not all irrational numbers are surds. For instance, constants like π and e are irrational but not surds.

Examples of Irrational Numbers and Their Values:

- e (Euler's Number): 2.718281
- π (Pi): 3.1415926
- $\sqrt{2}$: 1.41421356
- $\sqrt{3}$: 1.7320508

Other square roots such as $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$, $\sqrt{11}$, $\sqrt{13}$, $\sqrt{17}$, and $\sqrt{19}$ also exhibit irrational properties.

The cube root of 47 and logarithmic values like $\log 35$ further exemplify irrational numbers.

Example 6: $(\sqrt{2} + \sqrt{3})^2$ is a?**Solution:**

Given:

$(\sqrt{2} + \sqrt{3})^2$ we have to find the types of number

Formula Used:

$$(a + b)^2 = a^2 + 2ab + b^2$$

Calculation:

$$(\sqrt{2} + \sqrt{3})^2$$

$$\Rightarrow (\sqrt{2})^2 + 2 \times \sqrt{2} \times \sqrt{3} + (\sqrt{3})^2$$

$$\Rightarrow 2 + 2\sqrt{6} + 3$$

$$\Rightarrow 5 + 2\sqrt{6}$$

$\sqrt{6}$ is an irrational number, so, $(5 + 2\sqrt{6})$ is also an irrational number.

$\therefore$ The given number is an Irrational number.

Integers:

The set of all natural numbers, zero and negative of all natural numbers are called integers and it is denoted by I and Z. Ex;

$\{\dots -4, -3, -2, -1, 0, 1, 2, \dots\}$

- Negative integer is denoted by $I^- = \{-4, -3, -2, -1\}$
 - A positive integer is denoted by $I^+ = \{1, 2, 3, 4, \dots\}$
 - '0' is a non-negative integer as well as a non-positive integer.
-

Fractions:

Fractions represent a part of a whole or any number of equal parts. When spoken in terms of division, the top number (numerator) is divided by the bottom number (denominator).

- Proper fractions example: $\frac{3}{4}$ where the numerator is less than the denominator.
- Improper fractions example: $\frac{5}{3}$ where the numerator is greater than the denominator.
- Mixed numbers example: $2\frac{1}{2}$ which represents two whole parts and a half.

Example 7: Which among the following is a rational number equivalent to $\frac{3}{5}$

1. $\frac{9}{15}$
2. $\frac{9}{20}$
3. $\frac{9}{25}$
4. $\frac{3}{10}$

Solution:

Reduce the following fraction to its lowest number

Option 1

$$\frac{9}{15} = \frac{3}{5} \text{ (common multiple = 3)}$$

Option 2

$$\frac{9}{20} = \frac{9}{20} \text{ (no common multiple)}$$

Option 3

$$\frac{9}{25} = \frac{9}{25} \text{ (no common multiple)}$$

Option 4

$$\frac{3}{20} = \frac{3}{20} \text{ (no common multiple)}$$

∴ The correct answer is $\frac{9}{15}$.

Natural Numbers:

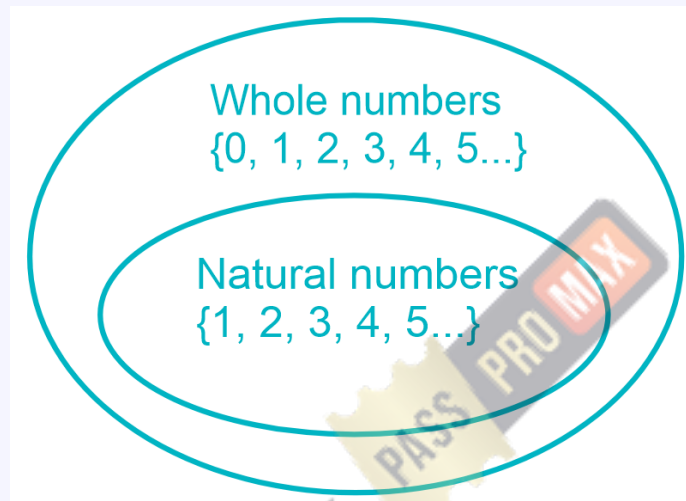
All counting numbers are called natural numbers and it is denoted by N. Ex; {1, 2, 3, 4...}

- The smallest natural number is 1 but there is no largest natural number.

Whole Numbers:

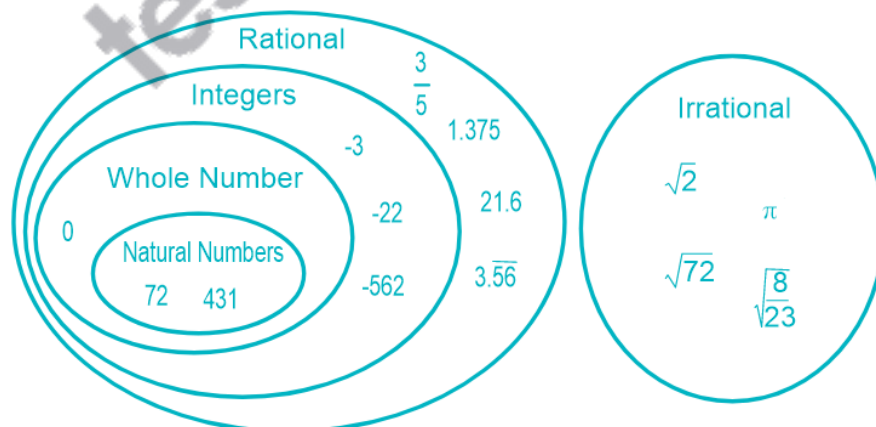
The set of all natural numbers together with zero are called whole numbers and it is denoted by W. Ex; {0, 1, 2, 3, 4...}

- A whole number is also called a “Non-negative Integer”.
- A whole number is also called a “Non-negative Integer”.



Real Number System

Examples of rational numbers, integers, whole numbers, natural numbers, and irrational numbers



Some special numbers:

Prime Number:

A number is called the prime number that has only two factors (1 and itself).
Ex; 2, 3, 5, 7, 11, 13.. Etc.

2 is the smallest even prime number

1 is neither prime nor composite

- Number of prime numbers between 1-50 = 15 prime numbers
 - Number of prime numbers between 51-100 = 10 prime numbers
 - Number of prime numbers between 1-100 = 25 prime numbers
-

Composite Number:

Those numbers are called composite numbers which have more than two factors. Ex; 4 because the factor of 4 is 1, 2 and 4.

- 4 is the smallest composite number.
 - 9 is the smallest odd composite number.
-

Co-Prime Number:

A pair of any two integers is called a Co-Prime Number, if their H.C.F is 1. Ex; (7, 11), (5, 11) etc.

Twin Prime Number:

A pair of two prime numbers is called a twin prime number, if the difference between these numbers is 2. Ex; (3, 5), (5, 7), (11, 13)etc.

Note:- (7, 11) is not a twin prime number.

Even and Odd Numbers:

Integers that are divisible by 2 are called even numbers and that are not divisible by 2 are known as odd numbers.

Ex:- ..., -4, -2, 0, 2, 4,... etc. are even integers also denotes the form $2K$ (where value of K varies from 1 to infinite)

And ..., -3, -1, 1, 3, 5, etc. are odd integers that also denote the form $2K \pm 1$ (where the value of K varies from 1 to infinite).

Perfect Numbers:

Perfect numbers are equal to the sum of their positive divisors, excluding themselves. They are rare and possess unique properties in number theory.

Example: 6, as $1 + 2 + 3 = 6$

Terminating and Non-Terminating Decimals

Terminating Decimals:

Decimals that have a finite number of digits after the decimal point.

- **Example:** 0.5, 3.25, -1.75
 - **Properties:** All terminating decimals are rational numbers.
-

Non-Terminating Decimals:

Decimals that continue infinitely without ending.

Types:

- **Repeating:** Decimal has a repeating sequence of digits (e.g., 0.333... or 0.142857...).
- **Non-Repeating:** Decimal has no repeating pattern (e.g., $\pi = 3.14159...$)

Example 8: The rational number with recurring decimal representation $0.\overline{124}$ is:

Solution:

Calculation:

$$\text{Let } x = 0.\overline{124}$$

Multiply by 10 on both sides

$$\Rightarrow 10x = 1.2424\ldots \quad \text{----(1)}$$

Multiply by 100 on both sides

$$\Rightarrow 1000x = 124.2424\ldots \quad \text{----(2)}$$

Subtract (1) from (2)

$$\Rightarrow 1000x - 10x = 124.2424 - 1.2424$$

$$\Rightarrow 990x = 123$$

$$\Rightarrow x = 123/990$$

$$\Rightarrow x = 41/330$$

$\therefore$ The correct answer is $41/330$.

Real-Life Examples and Applications:

Problem 1: Jack has a collection of irrational numbers: $\sqrt{3}$, $\sqrt{2}$, $\sqrt{6}$, $\sqrt{10}$, and $\sqrt{5}$. He needs to find the one closest to and not less than 1. After comparing their values, $\sqrt{2}$, being closest to 1, is identified as the ideal choice.

Problem 2: During a dice game, Alex rolls a five and must collect all the irrational numbers from a set including e , -5 , $\sqrt{9}$, $\sqrt{17}$, and π . Rational numbers like -5 , $\sqrt{9}$ (a perfect square), and $-2/8$ (a simple fraction) are excluded. Hence, Alex collects e , $\sqrt{17}$, and π as irrational numbers.

Conclusion: Irrational numbers outnumber rational ones and are infinitely numerous, making them impossible to fully list. Understanding the properties of irrational numbers, such as their non-repeating, endless decimal expansions and relationships with prime numbers, is crucial for identifying and working with them across various mathematical and practical applications.

Examples:

Example 9: Consider the following statements:

1. Every irrational number is a real number.
2. Every real number is a rational number.
3. Every rational number is a real number.
4. Every integer is a real number.

Which of the above statements are correct?

Solution:

Explanation:

Every rational number is a real number

- A rational number is a type of rational number. It can represent the p/q form, but q is not equal to zero.
- The numerator is an integer but the denominator is not equal to zero.
- It is a repeating decimal or terminating decimal.
- Example – $1/3$, $4/5$, $12/7$ etc

Every integer is a real number

- We know that an integer is a number that can be positive, negative and zero.
- It is also in fraction form
- The whole number are 0, 1, 2, 3,
- The integers do not include decimal numbers.

Real number – It is a number containing both rational and irrational numbers.

An irrational number is a number that can not represent the p/q form.
Non-terminating and non-repeating numbers.

Example – π , $\sqrt{11}$ etc.

$\therefore$ The correct answer is Statements 1,3 and 4 are correct.

Example 10: Which of the following is NOT true with reference to a real number?

1. All irrational numbers are real numbers
2. A real number need not necessarily be a point on the number line
3. A Real number is unique
4. All rational numbers are real numbers

Solution:

Concept Used:

The real numbers are classified as rational and irrational numbers.

Every real number is unique and can be represented in number lines.

Conclusion:

According to this concept,

Option 2 is wrong because we can represent all the real numbers in the number line.

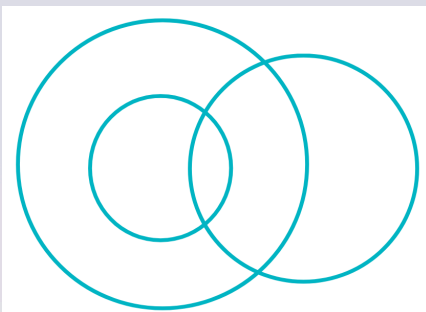
∴ The correct answer is option 2.

Example 12: The diagram that represent rational numbers, irrational number and real number is:

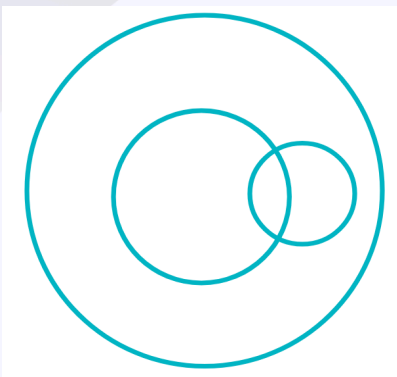


1.

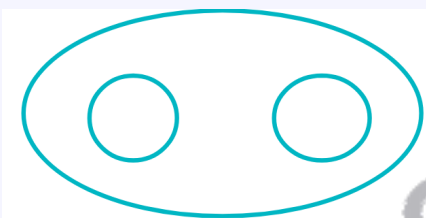
2.



3.



4.



Solution:



Option 4 represents the number line diagram that includes all rational and irrational numbers, as well as real numbers. It shows the rational numbers as discrete points on the line, and the irrational numbers as the infinite non-repeating decimals between the rational numbers. The diagram also includes irrational numbers that cannot be expressed as decimals, such as the square root of 2.

∴ The correct answer is option 4.

Example 13: Consider these statements

Statement 1: The difference between two negative integers cannot be a positive integer.

Statement 2: when a negative integer is divided by another negative integer then the resultant is a positive rational number

Solution:

Concept used:

Integers are non-fraction numbers & include both negative and positive numbers including zero.

$$(-m) - (-n) = n - m$$

$m, n = +ve$ integers

Statement 1: The difference between two negative integers cannot be a positive integer.

Let's take two positive integers as $m = 5$ and $n = 10$

$$\text{Now, } (-m) - (-n) = (-5) - (-10) = 10 - 5 = +5$$

On interchanging the values,

$$m = 10 \text{ and } n = 5$$

$$(-10) - (-5) = 5 - 10 = -5$$

With this, the addition of 2 negative integers can result in positive or negative integers depending on the values.

Therefore the given statement is False.

Statement 2: when a negative integer is divided by another negative integer then the resultant is a positive rational number

let us take two integers $-m, -n$

$$\Rightarrow -m/-n \Rightarrow 1 \text{ positive rational number}$$

$\therefore$ The correct answer is Statement 2 is correct & Statement 1 is incorrect.
